ENGINEERING BRIEF · AGENT C · FEATURE 2

Roomio — Persistence & Sharing

Turn Roomio from "a thing you play with once" into "a thing you save and send." Trustworthy autosave with a visible status, a clean my-designs library, and friendly sharing — a view-only showcase link to show off a room, plus exports (image, shopping list, floor-plan PDF). UX-first, grounded in proven save/share patterns.

AUTOSAVE + TRUST | SHOWCASE LINKS | EXPORTS

Repo: `~/Desktop/personal/roomio` · **Your folder:** `/persistence`
Coordination: `/shared` + `roomio.txt` + `/shared/LEARNINGS.md` · **Agent D merges**
Date: June 29, 2026 · You own the whole-house data model — ideal fit for persistence

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1. Why You & Why This Matters

> You are **Agent C**, returning for a second feature. You already built the multi-room house model — the whole-dwelling data structure (house → rooms → interiors → connectors). That makes you the natural owner of **persistence**: saving, loading, and sharing are fundamentally about serializing the entire structure you designed. You have the freshest mental model of the full data shape.

Why this is high-value: right now a Roomio design evaporates when the tab closes. Persistence and sharing are what turn it into something people return to and send to friends — the difference between a toy and a product. And this is the feature where UX *is* the product: save and sharing are exactly where users decide whether a tool feels **trustworthy** or scary. Lose someone's design once, or expose their editor when they meant to share a picture, and they don't come back.

ONBOARD FIRST (YOU KNOW THE DRILL)
Read `roomio.txt` + skim `/shared/LEARNINGS.md` for current state and the schemas A, C, and E publish. Post an onboarding entry. Code lives in `/persistence`; commit only your own paths; **Agent D merges**. This feature must serialize the FULL scene: house/rooms/connectors (yours), interiors + presets (A), lighting + time-of-day state (E). Coordinate their schemas via `/shared`.

2. Three Parts: Save, Library, Share/Export

#	Part	What it delivers
S1	Saving	Trustworthy autosave with a visible "Saving… / Saved" status, optimistic UI, a manual-save checkpoint, and unsaved-exit protection. Never lose a user's work.
S2	My Designs library	A home screen of saved designs with thumbnails, names, last-edited times; open, rename, duplicate, delete. Where designs live between sessions.
S3	Share & export	A copy-link share model (view vs edit), a dedicated view-only *showcase* link, and exports: image/snapshot, a furniture shopping list, and a floor-plan PDF.

STORAGE APPROACH — DECIDE EARLY, COORDINATE WITH D

Two viable tiers: (a) **local-first** (designs persist on-device + export/import a file) — fastest to ship, no accounts, no backend; or (b) **cloud accounts** (sign-in, designs sync, links resolve server-side) — needed for true cross-device + shareable URLs. Recommended path: ship local-first persistence + file export/import for S1/S2 immediately, and treat cloud accounts + live share URLs as a clearly-scoped follow-on once a backend exists. Note the decision in `roomio.txt`; it affects what "share link" can mean.

3. Part 1 — Saving (trust through feedback)

The documented best practice across Google Docs, Figma, Notion, Office: **autosave, but make it visible.** Silent saving creates anxiety; visible saving creates trust.

The save model

- **Autosave by default,** debounced — save shortly after the user pauses editing (e.g. ~1–3s of inactivity) rather than on every keystroke/drag, to avoid thrashing.
- **Visible status indicator** with two clear states: "Saving…" (with a spinner, while a save is in flight) and "Saved" (with a timestamp, e.g. "Saved just now" / "Saved 2 min ago"). This single element is what reassures users their work is safe.
- **Optimistic UI:** reflect the change in the scene immediately; persist in the background. The edit feels instant; the indicator shows the save catching up.
- **Manual save checkpoint:** keep a Save action (and honor Ctrl/Cmd-S) even with autosave on — heavy users reflexively hit it, and it doubles as a "make a named version" moment. Pressing it should give clear "Saved" feedback.
- **Unsaved-exit protection:** if a save is still pending when the user tries to close/navigate, warn them (beforeunload) so nothing is lost.

Error handling (trust depends on it)

- If a save fails, never silently drop it — show a clear "Couldn't save, retrying…" state and keep the user's data in memory; retry.
- Never autosave into an unrecoverable state — keep enough history that a bad edit can be undone (see versions below).

OPTIONAL BUT HIGH-TRUST: VERSION HISTORY
Keep lightweight snapshots (named manual saves, or periodic autosnapshots) so a user can roll back to "yesterday's layout." Even a simple list of restore points massively increases confidence to experiment. Scope as a stretch on top of basic save.

4. Part 2 — The "My Designs" Library

Where saved designs live between sessions — the home base users return to. Keep it visual and low-friction.

What it shows & does

- **A grid of design cards,** each with a thumbnail (a rendered snapshot of the room/house), a name, and a last-edited timestamp. Visual recognition beats text lists for spatial designs.
- **Core actions per design:** open, rename, duplicate ("use this as a starting point"), delete (with an undo or confirm so it's not a trap).
- **New design** entry that flows into the existing builder (blank wizard or a persona preset).
- **Sort/filter** by recent / name once there are many designs.
- Auto-generate the thumbnail from the scene on save so the library always looks current.

NAMING & FIRST-SAVE UX

Don't block creation on naming — let a design exist as "Untitled room" and autosave immediately, with an easy inline rename. Forcing a name-and-save dialog up front is friction users resent. (Roomio's current canvas already shows "Untitled room" — build on that.)

5. Part 3 — Sharing & Exports

Two distinct user intents, and the UX must separate them: "let someone *see* my room" vs "let someone *edit* it with me." Conflating them is the classic sharing mistake.

The share model (copy-link, clear access)

- **A Share button** opens a simple panel with a copy-link control and an explicit access choice: **can view** vs **can edit**. Default to the safer option (view).
- **A dedicated view-only "showcase" link** — the most important insight. People mostly want to *show off* a finished room without giving access to the editor or their other designs. A showcase link opens a clean, read-only walkthrough of just that design (ideally leveraging Agent B's flythrough/first-person view), never the editor. This is the Roomio equivalent of Figma's prototype-only link, and it's what users actually reach for.
- **Clarity over power:** show in plain language who can do what ("Anyone with this link can view"). Avoid exposing the editor or the user's library through a view link — the documented failure mode is a viewer escalating into the working file.
- If cloud accounts aren't built yet, "share" can start as **export/import a design file** (download a `.roomio` file, open it elsewhere) and a static showcase export, upgrading to live URLs once a backend exists. Note this in `roomio.txt`.

Exports (the take-aways that make Roomio useful beyond the app)

Export	What & why
Image / snapshot	A rendered still of the room/house to save or post. The shareable artifact people want most.
Walkthrough video	If Agent B's flythrough exists, export it as the shareable clip (coordinate with B; don't rebuild it).
Shopping / furniture list	A list of the archetypes placed (type, color, size, count) — turns a design into an actionable "what I need to buy." High practical value.
Floor-plan PDF	A top-down plan of the house with room labels + dimensions (you own the house model, so you have this data) — printable/shareable.

6. UX Principles (the heart of this)

This feature succeeds or fails on UX, not on storage mechanics. Hold to these:

- **Never lose work.** Autosave + visible status + unsaved-exit warning + save-failure retry. This is the trust foundation; everything else is secondary.
- **Make saving visible, not silent.** The "Saving… / Saved just now" indicator is small but does the heavy lifting of reassurance. Don't hide it.
- **Make edits feel instant.** Optimistic UI — show the result immediately, persist in the background.
- **Don't force decisions up front.** "Untitled room" + autosave beats a name-and-save dialog. Rename later, inline.
- **Separate "view" from "edit" sharing,** and default to the safe one. A showcase link must never expose the editor or other designs.
- **Plain-language access.** Tell users exactly who can do what with a shared link, in words, not icons alone.
- **Reversibility.** Delete should be undoable or confirmed; destructive actions never one-click-irreversible.
- **Respect the existing look.** Match Roomio's current clean panel UI (left controls, 3D canvas). Persistence UI should feel native, not bolted on.

THE TWO CARDINAL SINS TO AVOID

(1) Silently losing a design — the fastest way to lose a user forever. (2) A "share" that accidentally exposes the editor or the user's whole library when they only meant to show one room. Design defensively against both.

7. Data, Schema & Coordination

Persistence serializes the *entire* scene, so your save format must compose every agent's piece. You own the envelope; each agent owns its slice.

```
{
  "design_id": "d_001",
  "name": "Untitled room",
  "updated_at": "2026-06-29T22:10:00Z",
  "thumbnail": "<data-url or ref>",
  "version": 3,                          // for version history
  "house": { /* your house/rooms/connectors model */ },
  "interiors": { /* A: walls, materials, placed archetypes, preset id */ },
  "lighting": { /* E: lights, time_of_day, north_offset */ },
  "share": {
    "view_link_id": null,                // showcase (read-only)
    "edit_link_id": null,
    "access": "private"                  // private | view | edit
  }
}
```

Coordination (same protocol as the rest of the team)

- Code in `/persistence`; only other writable places are `/shared`, `roomio.txt`, `LEARNINGS.md`. Commit only your own paths — **Agent D merges**.
- Read `roomio.txt` + `LEARNINGS.md` at the start of every cycle. Watch for `[USER -> AGENT-C]` as top priority.
- This feature must compose every agent's schema — post `REQUEST -> AGENT-A` (serialize interiors + preset id), `REQUEST -> AGENT-E` (serialize lighting/time state), `REQUEST -> AGENT-B` (reuse the flythrough for the showcase link + video export). Publish the save-envelope schema to `/shared` so each agent confirms its slice round-trips.
- Treat `ISSUE -> AGENT-C` from D as high priority. Record any storage/format gotchas in `LEARNINGS.md`.
- **Backward compatibility:** older single-room saves (and the one-room-house wrapper) must still load. Version the envelope and migrate forward.

NO BROWSER-STORAGE TRAP IN ARTIFACTS
If any part renders inside a constrained artifact/preview context, remember localStorage/sessionStorage may be unavailable there — keep state in memory or the app's real persistence layer. In the actual app build, normal device/cloud storage is fine. Record the chosen mechanism in LEARNINGS.md.

8. Functional Requirements & Acceptance

ID	Requirement	Part
PS-1	Designs autosave (debounced) after edits; no manual save required to be safe.	S1
PS-2	Visible save status: "Saving…" (spinner) → "Saved <timestamp>".	S1
PS-3	Optimistic UI: edits show instantly, persist in background.	S1
PS-4	Manual save action + Ctrl/Cmd-S, with clear feedback.	S1
PS-5	Warn on exit with unsaved changes; retry on save failure, never silently drop.	S1
PS-6	"My Designs" library: thumbnail + name + last-edited; open/rename/duplicate/ delete.	S2
PS-7	New designs start as "Untitled room", autosaved, renamable inline.	S2
PS-8	Delete is undoable or confirmed.	S2
PS-9	Share panel with copy-link + explicit view/edit access, defaulting to view.	S3
PS-10	Dedicated view-only showcase link that opens a read-only walkthrough, never the editor or other designs.	S3
PS-11	Exports: image snapshot, furniture shopping list, floor-plan PDF (+ flythrough video if B's exists).	S3
PS-12	Save envelope composes house+interiors+lighting; older saves still load (versioned migration).	All
PS-13	Code in /persistence; coordinate via /shared+roomio.txt+LEARNINGS.md; D merges; no edits to others' source.	Coord

Acceptance test (the "done" bar)

Furnish a room, make an edit, and watch the status go "Saving…" → "Saved just now" without touching a button; close and reopen the app and the design is exactly as left, listed in "My Designs" with a thumbnail. Rename it inline; duplicate it; delete one and undo the delete. Hit a simulated save failure and see a retry state, not silent loss. Open the Share panel, copy a view-only showcase link, open it in a fresh/incognito context, and confirm it shows a read-only walkthrough of just that room with no editor and no access to other designs. Export an image, a furniture shopping list, and a floor-plan PDF. Confirm an older single-room save still loads.

9. Milestones

Milestone	Deliverable	Definition of done
C2-0	Read roomio.txt + LEARNINGS.md; scaffold /persistence; decide storage tier (local-first vs cloud) with D; publish save-envelope schema; onboarding entry	Schema in /shared; storage decision logged; A/E/B pinged for their slices.
C2-1	S1: autosave + visible status + optimistic UI + manual save + unsaved-exit warning	Edits persist with "Saving…/Saved"; nothing lost on reload.
C2-2	S1: save-failure retry + (stretch) version history	Failures retried not dropped; optional restore points.
C2-3	S2: My Designs library with thumbnails + open/rename/duplicate/delete	Designs persist between sessions, visually browsable.
C2-4	S3: share panel + view/edit access + view-only showcase link	Showcase link opens read-only walkthrough, no editor leak.
C2-5	S3: exports — image, shopping list, floor-plan PDF (+ video via B)	All exports produce real downloadable artifacts.
C2-6	Backward-compat migration + polish + match Roomio's UI	Old saves load; UI feels native; cleanup.

One-line summary. In `/persistence` : trustworthy debounced autosave with a visible "Saving…/Saved" status, optimistic UI, manual checkpoint and unsaved-exit protection; a visual "My Designs" library (thumbnail, rename, duplicate, delete, "Untitled room" default); and friendly sharing — a view-only showcase link that never exposes the editor, plus image / shopping-list / floor-plan-PDF exports. Serialize the full scene (house+interiors+lighting), keep old saves loading, coordinate slices with A/E/B via /shared; Agent D merges.

End of brief · Roomio Persistence & Sharing (Agent C) · Never lose work; never leak the editor through a view link.